

PAC - chat

Given Data.

- VType (Vehicle type)
- CSOC (Current state of charge)
- RSOC (Required state of charge)
- PDuration (Parking duration)
- Time (Hour at arrival-24)
- Member
- Disabled
- Available.

Results Required.

- Notification. (Availability/Eligibility/Not charging needed/)
- Prior (Priority)
- Time S (Peak or offPeak)
- Cost (Net charging cost)
- Plost (Parking cost)
- TDiscount (Total Discount)
- TPay (Final Payment)
- Long Stay duration alert.

Processing Required

- evaluate charge needed.
 $RCharge = RSOC - CSOC$

- Set priority to emergency if $CSOC \leq 15$ AND $RSOC \geq 80$.
otherwise Normal.

- Determine time slot and Base Rate.
BRate $\Rightarrow 50$ for time 17:00 - 21:59
BRate = 35 for offpeak

- Calculate Discount Percentage (DPercentage) and discount
 $DAmt = RCost \times DPercentage$

- Calculate BPark according to duration.
- Parking free for disabled and 20% discount for Members
- $TPay \leftarrow (Cost + PCost)$

Alternative solution.

- if Available = 'N'. JACU Skip EV charge parking calculator only

- if VType = 1 $CSOC \geq 40$ and Not eligible for EV charging

- if $RCharge \leq 0$
No charging needed

- Emergency charging priority by passes Membership discounts on charging

- if PDuration > 8
Issue long-stay relocation warning.